We create chemistry

# Safety data sheet



## 1. Substance/preparation and manufacturer/supplier identification

**Product name:**
Basoflux® RD 5119 T

Use: Performance Chemicals for Oilfield Applications

Manufacturer/supplier:
BASF South East Asia Pte Ltd.
128 Beach Road #18-01
Guoco Midtown, 189773, Singapore
Telephone: +65 8322 4420
Telefax number: +65 6 334-0330
E-mail address: benny.zou@basf.com

Emergency information:
Singapore Emergency Toll-Free Number:
Telephone: 1800-723-1361
International emergency number:
Telephone: +49 180 2273-112

## 2. Hazard identification

Classification of the substance and mixture:
Flammable liquids: Cat.4
Specific target organ toxicity — single exposure: Cat.3 (May cause drowsiness and dizziness.)
Hazardous to the aquatic environment - acute: Cat.3
Hazardous to the aquatic environment - chronic: Cat.3

Label elements and precautionary statement:

Pictogram:

Signal Word:
Warning

Hazard Statement:

| | |
|---|---|
| H227 | Combustible liquid. |
| H336 | May cause drowsiness or dizziness. |
| H402 | Harmful to aquatic life. |
| H412 | Harmful to aquatic life with long lasting effects. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P271 | Use only outdoors or in a well-ventilated area. |
| P210 | Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. |
| P280 | Wear protective gloves and eye protection or face protection. |
| P273 | Avoid release to the environment. |
| P261 | Avoid breathing mist or vapour or spray. |

Precautionary Statements (Response):

| | |
|---|---|
| P312 | Call a POISON CENTER or physician if you feel unwell. |
| P304 + P340 | IF INHALED: Remove person to fresh air and keep comfortable for breathing. |
| P370 + P378 | In case of fire: Use foam or dry powder for extinction. |

Precautionary Statements (Storage):

| | |
|---|---|
| P403 + P233 | Store in a well-ventilated place. Keep container tightly closed. |
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Other hazards which do not result in classification:
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Repeated exposure may cause skin dryness or cracking. Product contains the following components and may cause carcinogenicity:
naphthalene

## 3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

Preparation based on: polyacrylate, solvent(s)

**Hazardous ingredients**

Distillates (petroleum), hydrotreated light
Content (W/W): >= 25 % - < 50 % Asp. Tox.: Cat. 1
CAS Number: 64742-47-8 Flam. Liq.: Cat. 4

solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Content (W/W): >= 20 % - < 25 % Asp. Tox.: Cat. 1
CAS Number: 64742-94-5 Flam. Liq.: Cat. 4
STOT SE: Cat. 3 (drowsiness and dizziness)
Aquatic Acute: Cat. 2
Aquatic Chronic: Cat. 2

1,2,4-trimethylbenzene
Content (W/W): >= 1 % - < 3 % Asp. Tox.: Cat. 1
CAS Number: 95-63-6 Flam. Liq.: Cat. 3
Acute Tox.: Cat. 4 (Inhalation - vapour)
Skin Irrit.: Cat. 2
Aquatic Chronic: Cat. 2
Eye Irrit.: Cat. 2A
STOT SE: Cat. 3 (irr. to respiratory syst.)
Aquatic Acute: Cat. 2

naphthalene (as a component of solvent naphtha heavy)
Content (W/W): > 0 % - < 0.3 % Flam. Sol.: Cat. 2
CAS Number: 91-20-3 Acute Tox.: Cat. 4 (oral)
Carc.: Cat. 2
Aquatic Acute: Cat. 1
Aquatic Chronic: Cat. 1
M-factor acute: 1
M-factor chronic: 1

## 4. First-Aid Measures

General advice:
Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

Suitable extinguishing media:
dry powder, foam

Specific hazards:
harmful vapours, carbon oxides, hydrocarbons
Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

## 6. Accidental Release Measures

Personal precautions:
Use personal protective clothing.Breathing protection required.Avoid inhalation.Avoid contact with skin.Ensure adequate ventilation.Sources of ignition should be kept well clear.

Environmental precautions:
Contain contaminated water/firefighting water.Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

## 7. Handling and Storage

Handling

Ensure thorough ventilation of stores and work areas. Avoid skin contact. Avoid inhalation of the product. Wear suitable personal protective equipment.

No eating, drinking, smoking or tobacco use at the place of work. Remove contaminated clothing and protective equipment before entering eating areas. Wash hands before breaks and at end of work.

Protection against fire and explosion:
Sources of ignition should be kept well clear. Take precautionary measures against static discharges. If delivered in plastic packing, highest permissable emptying temperature is 5 Kelvin below the flash point.

Storage

Segregate from foods and animal feeds. Segregate from nitrites.
Further information on storage conditions: Keep container tightly closed and in a cool place.

Storage stability:
Storage temperature: <= 50 °C

The packed product is not damaged by low temperatures or by frost.
Protect from temperatures above: 50 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

## 8. Exposure controls and personal protection

Components with occupational exposure limits

1,2,4-trimethylbenzene, 95-63-6;
TWA value 10 ppm (ACGIHTLV)
TWA value 123 mg/m3 ; 25 ppm (OEL (SG))

Solvent naphtha (petroleum), heavy arom., 64742-94-5;
TWA value 200 mg/m3 (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Application restricted to conditions in which there are negligible aerosol exposures.
Skin Designation (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Danger of cutaneous absorption
Skin Designation (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Danger of cutaneous absorption
TWA value 200 mg/m3 (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Application restricted to conditions in which there are negligible aerosol exposures.

Distillates (petroleum), hydrotreated light, 64742-47-8;

TWA value 200 mg/m3 (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Application restricted to conditions in which there are negligible aerosol exposures.
Skin Designation (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Danger of cutaneous absorption

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Gas filter for gases/vapours of organic compounds (boiling point >65 °C, e. g. EN 14387 Type A)

Hand protection:
Chemical resistant protective gloves
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
fluoroelastomer (FKM) - 0.7 mm coating thickness
Polyethylene-Laminate (PE laminate) - ca. 0.1 mm coating thickness
Suitable materials for short-term contact (recommended: At least protective index 2, corresponding > 30 minutes of permeation time according to EN ISO 374-1)
nitrile rubber (NBR) - 0.4 mm coating thickness
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses with side-shields.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
Wearing of closed work clothing is required additionally to the stated personal protection equipment.
Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and Chemical Properties

| | |
|---|---|
| Form: | liquid to waxy |
| Colour: | colourless to yellowish |
| Odour: | mineral oil-like |
| Odour threshold: | not determined |
| pH value: | not applicable, insoluble |
| Melting point: | 17 - 23 °C |

| | | |
|---|---|---|
| Boiling point: | > 190 °C | |
| Flash point: | 62.5 °C<br>The product has not been tested.<br>The statement has been derived from substances/products of a similar structure or composition. | (ISO 2719) |
| Evaporation rate: | not determined | |
| Flammability (solid/gas): | Combustible liquid. | (derived from flash point) |
| Lower explosion limit: | For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point. | |
| Upper explosion limit: | For liquids not relevant for classification and labelling. | |
| Ignition temperature: | > 230 °C | (DIN 51794) |
| Thermal decomposition: | 375 °C , > 200 J/g<br>self-accelerating reaction | (DSC (DIN 51007)) |
| Self ignition: | not self-igniting | |
| Explosion hazard: | not explosive | |
| Fire promoting properties: | not fire-propagating | |
| Vapour pressure: | approx. 5.96 hPa<br>(20 °C)<br>static<br>approx. 17.82 hPa<br>(50 °C)<br>static<br>approx. 20.98 hPa<br>(55 °C)<br>static | |
| Density: | 0.89 g/cm3<br>(15 °C) | (ISO 2811-3) |
| | 0.879 g/cm3<br>(20 °C) | (ISO 2811-3) |
| | 0.8581 g/cm3<br>(50 °C) | (ISO 2811-3) |
| Relative density: | No data available. | |
| Relative vapour density (air): | > 1<br>(20 °C)<br>Heavier than air. | (estimated) |
| Solubility in water: | dispersible when hot | |
| Hygroscopy: | Non-hygroscopic | |
| Solubility (qualitative) solvent(s): | aromatic hydrocarbons<br>soluble | |

Partitioning coefficient n-octanol/water (log Pow):
not applicable for mixtures

| | | |
|---|---|---|
| Viscosity, dynamic: | 932 mPa.s<br>(23 °C) | |
| Viscosity, kinematic: | approx. 1,060 mm2/s<br>(23 °C) | (calculated (from dynamic viscosity)) |

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

Specific Surface Area:
No data available.

Particle Shape:
No data available.

Dustiness:
No data available.

## 10. Stability and Reactivity

Conditions to avoid:
Avoid all sources of ignition: heat, sparks, open flame. Avoid electro-static discharge.

Thermal decomposition: 375 °C, > 200 J/g (DSC (DIN 51007))
self-accelerating reaction

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Corrosion to metals: No corrosive effect on metal.

Hazardous reactions:
No hazardous reactions when stored and handled according to instructions.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Routes of exposure

#### Acute oral toxicity

Experimental/calculated data:

(oral):No data available.

**Acute oral toxicity**

ATE (oral): > 5,000 mg/kg

**Acute inhalation toxicity**

(by inhalation):No data available.

**Acute dermal toxicity**

(dermal):No data available.

ATE (dermal): > 5,000 mg/kg

**Assessment of acute toxicity**

No data available.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified

**Acute oral toxicity**

Experimental/calculated data:
LD50 rat (oral): 6,318 mg/kg (OECD Guideline 401)

Information on: Distillates (petroleum), hydrotreated light

**Acute oral toxicity**

Experimental/calculated data:
LD50 rat (oral): > 5,000 mg/kg (OECD Guideline 420)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.
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Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified

**Acute inhalation toxicity**

Experimental/calculated data:
LC50 rat (by inhalation): > 4.688 mg/l (OECD Guideline 403)

Information on: Distillates (petroleum), hydrotreated light

**Acute inhalation toxicity**

Experimental/calculated data:
LC50 rat (by inhalation): > 5.28 mg/l 4 h (OECD Guideline 403)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The vapour was tested.
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Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified

**Acute dermal toxicity**

Experimental/calculated data:
LD50 rabbit (dermal): > 2,000 mg/kg (OECD Guideline 402)

Information on: Distillates (petroleum), hydrotreated light

**Acute dermal toxicity**

Experimental/calculated data:
LD50 rabbit (dermal): > 2,000 mg/kg (OECD Guideline 402)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.
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**Symptoms**

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far

## Irritation

Assessment of irritating effects:
May cause slight irritation to the skin. May cause slight irritation to the eyes. May cause slight irritation to the respiratory tract.

Experimental/calculated data:
Skin corrosion/irritation:No data available.

Serious eye damage/irritation:No data available.

## Respiratory/Skin sensitization

Assessment of sensitization:
Based on available data, the classification criteria are not met.

Experimental/calculated data:
No data available.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of sensitization:
No sensitizing effect.

Information on: Distillates (petroleum), hydrotreated light
Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.
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## Germ cell mutagenicity

Assessment of mutagenicity:
Based on the ingredients, there is no suspicion of a mutagenic effect.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of mutagenicity:
Based on available data, the classification criteria are not met.

Information on: Distillates (petroleum), hydrotreated light
Assessment of mutagenicity:
In the majority of tests performed (bacteria/microorganisms/cell cultures) a mutagenic effect was not found. A mutagenic effect was also not observed in in-vivo assays. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.
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## Carcinogenicity

Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Experimental/calculated data:
No data available.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Information on: Distillates (petroleum), hydrotreated light
Assessment of carcinogenicity:
Long-term exposure to highly irritating concentrations resulted in skin tumors in animals. A carcinogenic effect in humans can be excluded after brief skin contact. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.
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## Reproductive toxicity

Assessment of reproduction toxicity:
Based on the ingredients, there is no suspicion of a toxic effect on reproduction.

Experimental/calculated data:
No data available.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of reproduction toxicity:
Based on available data, the classification criteria are not met.

Information on: Distillates (petroleum), hydrotreated light
Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.
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## Developmental toxicity

Assessment of teratogenicity:
Based on the ingredients, there is no suspicion of a teratogenic effect.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of teratogenicity:
Based on available data, the classification criteria are not met.

Information on: Distillates (petroleum), hydrotreated light
Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.
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### Specific target organ toxicity (single exposure)

Possible narcotic effects (drowsiness or dizziness).

### Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Experimental/calculated data:
No data available.

### Aspiration hazard

No aspiration hazard expected.

### Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components. Has a degreasing effect on skin.

## 12. Ecological Information

### Ecotoxicity

Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. Harmful to aquatic life with long lasting effects.

Toxicity to fish:
No data available.

Aquatic invertebrates:
No data available.

Aquatic plants:
No data available.

Microorganisms/Effect on activated sludge:
No data available.

Chronic toxicity to fish:
No data available.

Chronic toxicity to aquatic invertebrates:
No data available.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Toxicity to fish:
LL50 (96 h) 2 mg/l, Oncorhynchus mykiss

LL50 (96 h) < 5 mg/l, Oncorhynchus mykiss
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Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Aquatic invertebrates:
EL50 (48 h) 3 mg/l, Daphnia magna

EL50 (48 h) < 10 mg/l, Daphnia magna
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Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Chronic toxicity to fish:
NOEL (28 d) 0.441 mg/l, Oncorhynchus mykiss
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Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Chronic toxicity to aquatic invertebrates:
NOEL (21 d), 0.771 mg/l, Daphnia magna
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### Mobility

Assessment transport between environmental compartments:
No data available.

### Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

### Bioaccumulation potential

Assessment bioaccumulation potential:
Accumulation in organisms is expected.

Information on: Distillates (petroleum), hydrotreated light
Assessment bioaccumulation potential:
Accumulation in organisms is expected.
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### Additional information

Other ecotoxicological advice:
The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components. Do not discharge product into the environment without control.

## 13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.
Do not discharge into waterways or sewer systems

Contaminated packaging:
Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

## 14. Transport Information

**Domestic transport:**

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Sea transport**
IMDG

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable<br>Marine pollutant: no |
| Special precautions for user | None known |

**Air transport**
IATA/ICAO

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number | Not applicable |
| Proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

### Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## 15. Regulatory Information

**Other regulations**

## 16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.